Exp No: 8	NMAP to Discover Live Hosts using Nmap Scans (ICMP, TCP / UDP) on the TryHackMe Platform

AIM:

To use the Nmap tool on the TryHackMe platform to discover a live host using ICMP, TCP and UDP ping scans and to find the open ports of the host.

ALGORITHM / PROCEDURE:

1. Log in to the TryHackMe website, open the *Nmap* room, click *Start AttackBox* and deploy the lab machine. Note the IP address of the AttackBox (10.48.191.226) and of the target machine (10.48.191.35).
2. Check the target with `ping -c 4 10.48.191.35`. All 4 packets are lost, so the target does not answer the ICMP ping.
3. ICMP echo scan: run `sudo nmap -PE 10.48.191.35` to send ICMP echo requests.
4. TCP SYN ping: run `sudo nmap -PS80 10.48.191.35` to send a TCP SYN packet to port 80.
5. UDP ping: run `sudo nmap -PU53 10.48.191.35` to send a UDP packet to port 53.
6. Run `nmap 10.48.191.226` on the AttackBox itself to see its own open ports.
7. Note the host status and the open ports reported by every scan and compare the results.

Commands used:

Scan type	Option	Command
ICMP ping (check)	`ping`	`ping -c 4 10.48.191.35`
ICMP echo ping	`-PE`	`sudo nmap -PE 10.48.191.35`
TCP SYN ping	`-PS`	`sudo nmap -PS80 10.48.191.35`
UDP ping	`-PU`	`sudo nmap -PU53 10.48.191.35`
Default scan of the AttackBox	`-`	`nmap 10.48.191.226`

DIAGRAM:

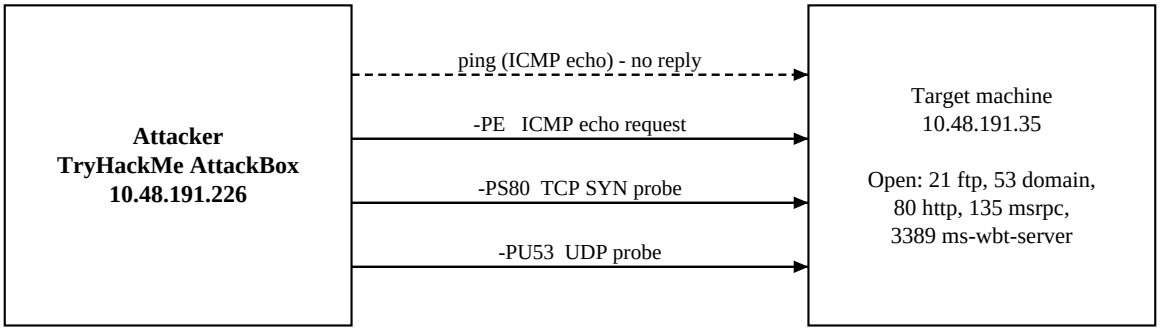


Fig 1: Host discovery probes sent from the AttackBox to the target

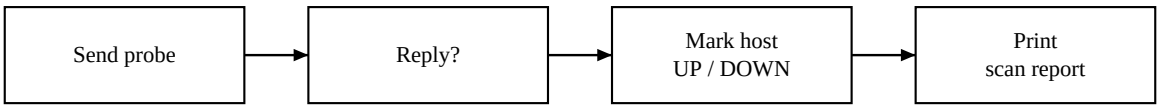
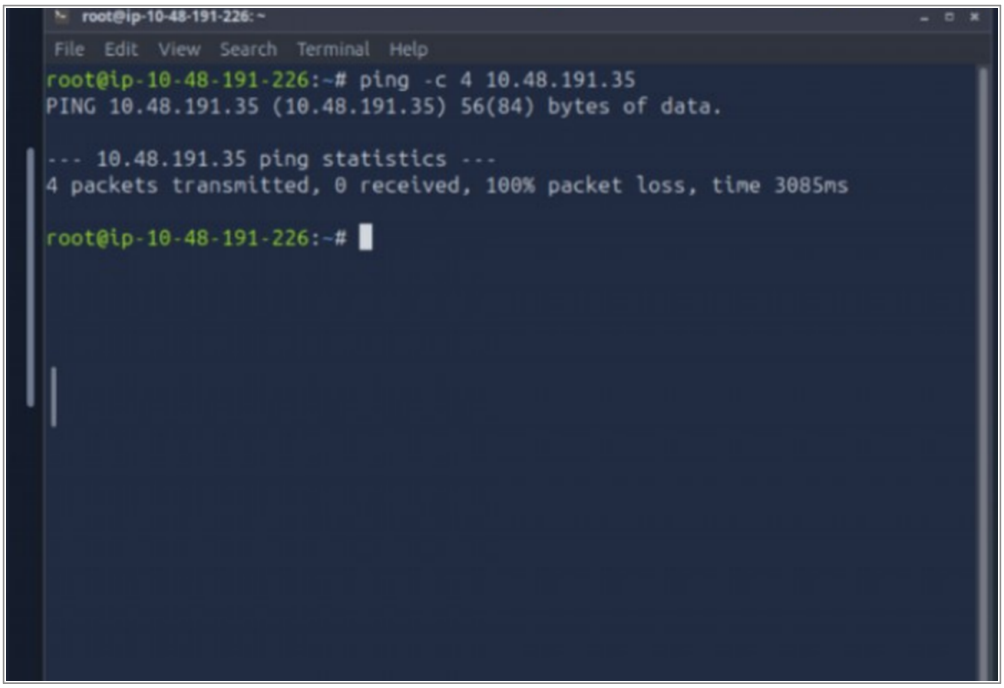
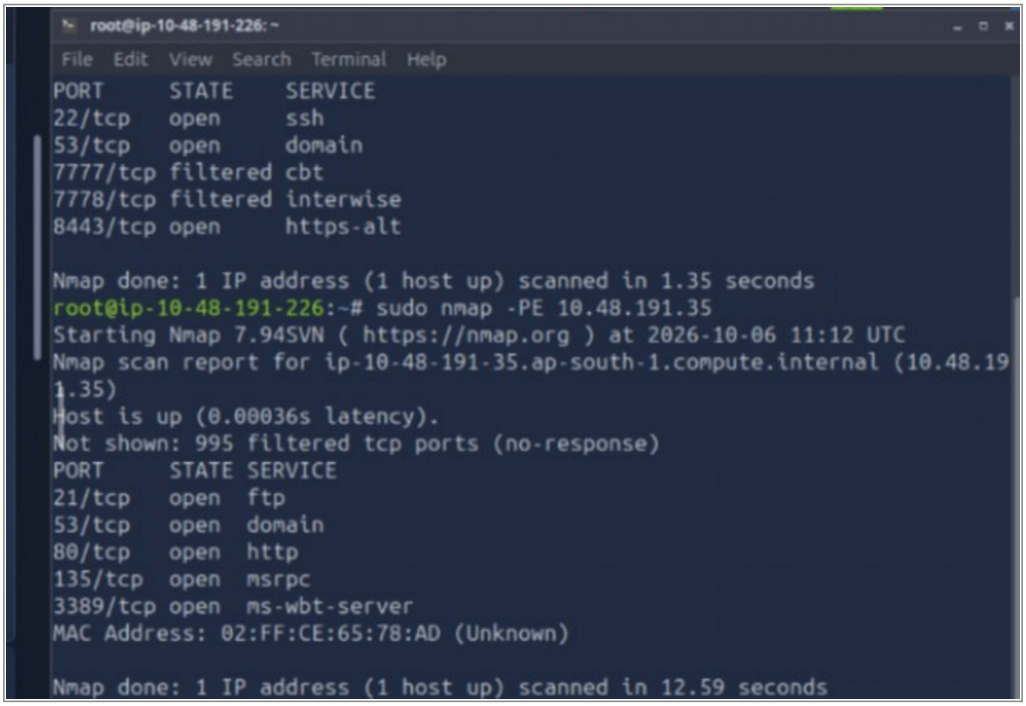


Fig 2: Working of a host discovery scan

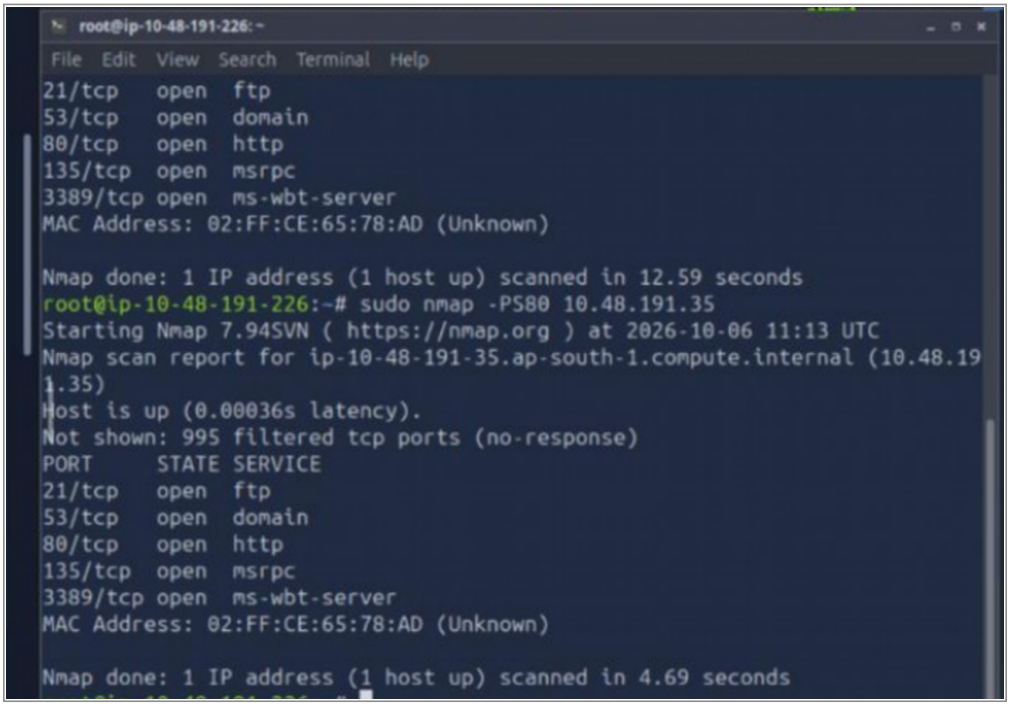
OUTPUT:



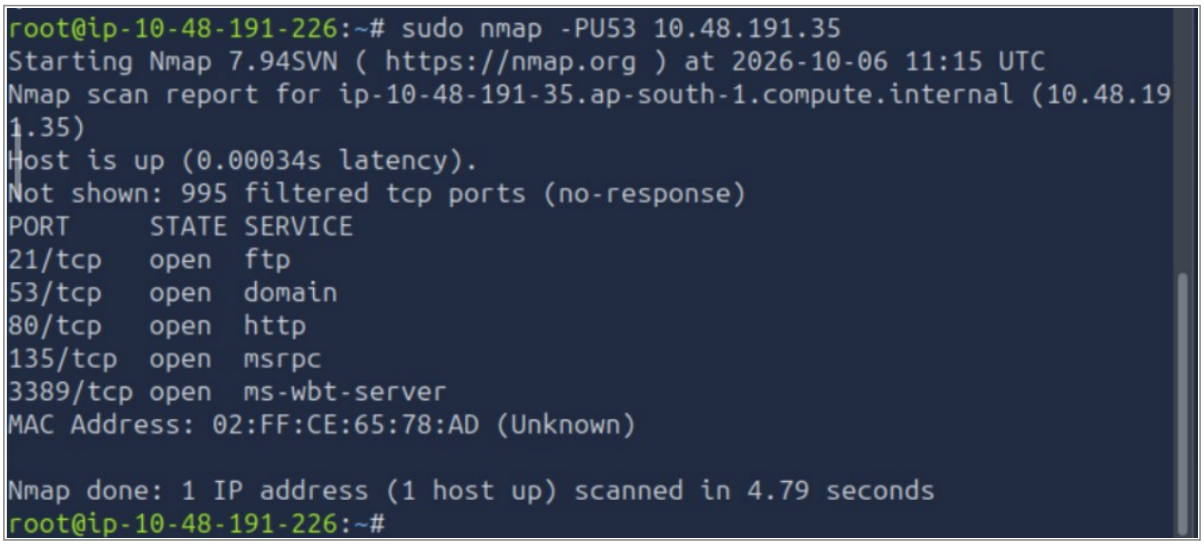
ping -c 4 10.48.191.35 : 4 packets transmitted, 0 received, 100% packet loss



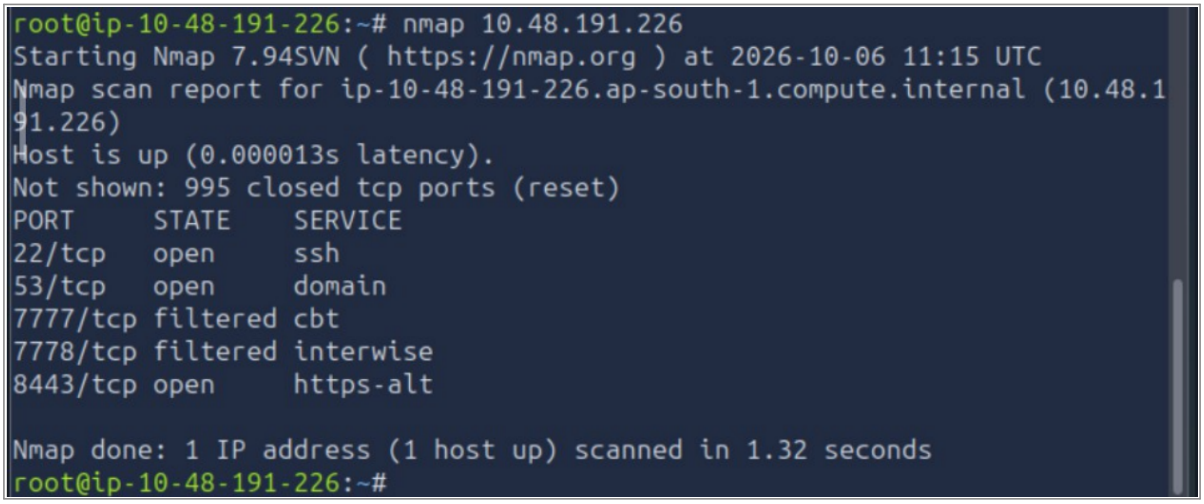
sudo nmap -PE 10.48.191.35 (ICMP echo scan)



sudo nmap -PS80 10.48.191.35 (TCP SYN ping)



sudo nmap -PU53 10.48.191.35 (UDP ping)



nmap 10.48.191.226 (scan of the AttackBox)

RESULT:

Thus the live host was discovered using Nmap on the TryHackMe platform. The normal ping got 100% packet loss, but the Nmap ICMP, TCP SYN and UDP scans reported the target 10.48.191.35 as up (its MAC address was shown, so it is on the same subnet) and listed the open ports 21 (ftp), 53 (domain), 80 (http), 135 (msrpc) and 3389 (ms-wbt-server) with the other 995 ports filtered.